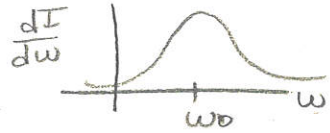
* bras of particles at finite temp $k_B T$ emitting at a normal frequency ω_0

* derive why we see $\omega \neq \omega_0$ for far away from the gas particles (get a spread of frequencies around ω_0)



* <u>Define W</u> = probability to emit a photon per unit of time in the rest frame

- $$W = \frac{\text{Probability}}{\text{time}} = \frac{dN}{dt'} \implies dN = W\,dt'$$

re-write W in terms of the # of particles dN after wait time dt into a frequency interval dω at solid angle dΩ in the rest frame of the atom, $v = 0$

- $$dN = \underline{\;?\;}\, dt'\, d\omega'\, d\Omega'$$ where $\frac{v}{c} \ll 1$ for particles to simplify

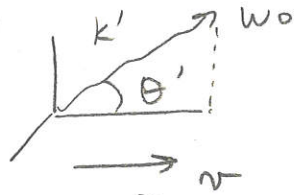
- $$dN = \frac{1}{4\pi} W\, \delta(\omega' - \omega_0)\, dt'\, d\Omega'\, d\omega'$$

note: if integrate over $d\Omega'$ & $d\omega'$, get original $dN = W dt'$ so these are the constants & terms needed.

also, $dN \neq dN'$ since # of particles will not change between rest & lab frame

* <u>Transform dω'</u>

lab frame



rest frame

$$k^\alpha = L^\alpha_\beta k_\alpha'$$

$$k^\alpha = (\omega/c, \vec{k})$$

$$\begin{pmatrix} \gamma & \beta\gamma & 0 & 0 \\ \beta\gamma & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \omega'/c \\ k'\cos\theta' \\ k_y' \\ k_z' \end{pmatrix} = \begin{pmatrix} \omega/c \\ k_x \\ k_y \\ k_z \end{pmatrix}$$

only care about ω/c term

$$\frac{\omega}{c} = \gamma\left(\frac{\omega'}{c}\right) + \beta\gamma k'\cos\theta' = \gamma\left(\frac{\omega'}{c} + \beta k'\cos\theta'\right)$$

but $\gamma \approx 1$ since $v/c \ll 1$

* so $\frac{\omega}{c} \approx \left(\frac{\omega'}{c} + \frac{v\omega'}{c^2}\cos\theta'\right)$ since $\gamma \approx 1$ & $k' = \frac{\omega'}{c}$

$$\omega = \omega' + \frac{v\omega'}{c}\cos\theta' \quad \text{solve for } \omega'$$

$$\omega' = \frac{\omega}{\left(1 + v/c\cos\theta'\right)}$$

* <u>Plug ω' into dN formula</u>

- $$dN = \frac{1}{4\pi} W \delta\left(\frac{\omega}{(1+v/c\cos\theta')} - \omega_0\right) dt'\, d\Omega' \left(\frac{d\omega}{1 + v/c\cos\theta'}\right)$$

Note: $\delta(ax) = \frac{1}{a}\delta(x)$

in our case: $a = \frac{1}{1+v/c\cos\theta'}$, so $1/a$ will cancel when factored out

- $$dN = \frac{1}{4\pi} W \cancel{(1+v/c\cos\theta')}\, \delta\left(\omega - \omega_0(1+v/c\cos\theta')\right) dt\, d\Omega' \frac{d\omega}{\cancel{(1+v/c\cos\theta')}}$$

& where $dt = \gamma dt'$ & $\gamma \approx 1$ so $dt = dt'$

- $$dN = \frac{1}{4\pi} W \delta\left(\omega - \omega_0(1 - v/c\cos\theta')\right) dt\, d\omega\, d\Omega'$$

* <u>Integrate over solid angle:</u>

- $d\Omega' = d\phi' d\theta' \sin\theta' \Rightarrow d\phi'(-d(\cos\theta')) = d\Omega'$

re-write in terms of $d\cos\theta'$ so can integrate δ function

- re-write δ function:

$$\delta\left(\omega - \omega_0 - \frac{\omega_0 v}{c}\cos\theta'\right) \Rightarrow \frac{c}{\omega_0 v}\delta\left(\frac{(\omega-\omega_0)c}{\omega_0 v} - \cos\theta'\right)$$

via $\delta(ax) = \frac{1}{a}\delta(x)$

- Therefore, have integral:

$$dN = -\frac{W}{4\pi}\frac{c}{\omega_0 v}\int_{\phi=0}^{2\pi}\int_{-1}^{1} \delta\left(\frac{(\omega-\omega_0)c}{\omega_0 v} - \cos\theta'\right) dt\, d\omega\, d\phi'\, d(\cos\theta')$$

where $\theta = 0 \Rightarrow \cos\theta = -1$ & $\theta = \pi \Rightarrow \cos\theta = 1$ from substitution

* $$dN = -\frac{W}{4\pi}\frac{c}{\omega_0 v}\int_{-1}^{+1}\delta\left(\frac{(\omega-\omega_0)c}{\omega_0 v} - \cos\theta'\right) dt\, d\omega\, d\cos(\theta')\,(2\pi)$$

The δ function will give a heavy-side function θ since there are many pairs of θ' & ω that can satisfy the delta function

$$dN = +\frac{W}{2}\frac{c}{\omega_0 v}\,\theta\left(1 - \frac{c}{\omega_0 v}|\omega-\omega_0|\right) dt\, d\omega$$

$$dN = \frac{Wc}{2\omega_0 v}\,\theta\left(1 - \frac{c}{\omega_0 v}|\omega-\omega_0|\right) dt\, d\omega$$

this dN describes the # of photons emitted by 1 particle

* <u>Apply a boltzmann distribution</u>

how many particles per unit of velocity space?

$$dn = D\, e^{-mv^2/2k_BT}\, v^2 dv$$

D = some const
dn = # of particles

combine dN & dn to get # of photons emitted by <u>all</u> particles $\Rightarrow dN_{total}$

$$dN_{tot} = \int dN\, dn$$

$$= \frac{WcD}{2\omega_0}\, dt \int v\, e^{-mv^2/2k_BT}\,\theta\left(1 - \frac{c}{\omega_0 v}|\omega-\omega_0|\right) dv\, d\omega$$

to integrate the heavy side function (which is a function of both v & ω), make the limits of dv such that we are only integrating over the region such that the heavy side function is the value of 1

$$dN_{tot} = \frac{WcD}{2\omega_0}\, dt \int_{\frac{c}{\omega_0}|\omega-\omega_0|}^{\infty} v\, e^{-mv^2/2k_BT}\, dv\, d\omega$$

integrate over dv^2 such that $d(v^2) = 2v\,dv$

$$dv = \frac{d(v^2)}{2v}$$

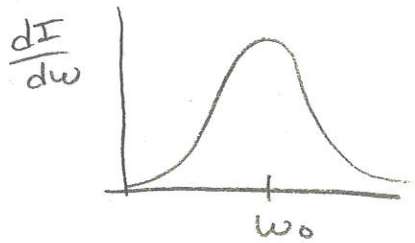
- $$dN_{tot} = \frac{WcD}{4\omega_0}\, dt \int_{\frac{c}{\omega_0}|\omega-\omega_0|}^{\infty} e^{-mv^2/2k_BT}\, dv^2\, d\omega$$

$$= \frac{WcD}{4\omega_0}\left(\frac{2k_BT}{m}\right) e^{-mv^2/2k_BT}\Big|_{\frac{c}{\omega_0}|\omega-\omega_0|}^{\infty} dt\, d\omega$$

$$\boxed{dN = \frac{WcDk_BT}{2\omega_0 m}\, e^{-\frac{mc^2(\omega-\omega_0)^2}{2k_BT\omega_0^2}}\, dt\, d\omega}$$

where $\omega - \omega_0 = \Delta\omega$, hence

$$dN \propto e^{\frac{-mc^2}{2k_BT}\left(\frac{\Delta\omega}{\omega_0}\right)^2}$$ which is a gaussian distribution



Therefore we have proved that we get a distribution of frequencies around ω_0 & not just a single frequency